

Day4:

CodeCommit-> CodeBuild (buildspec.yaml) -> CodeDeploy-> CodePipeline

buildspec.yaml => build

appspec.yaml => deploy

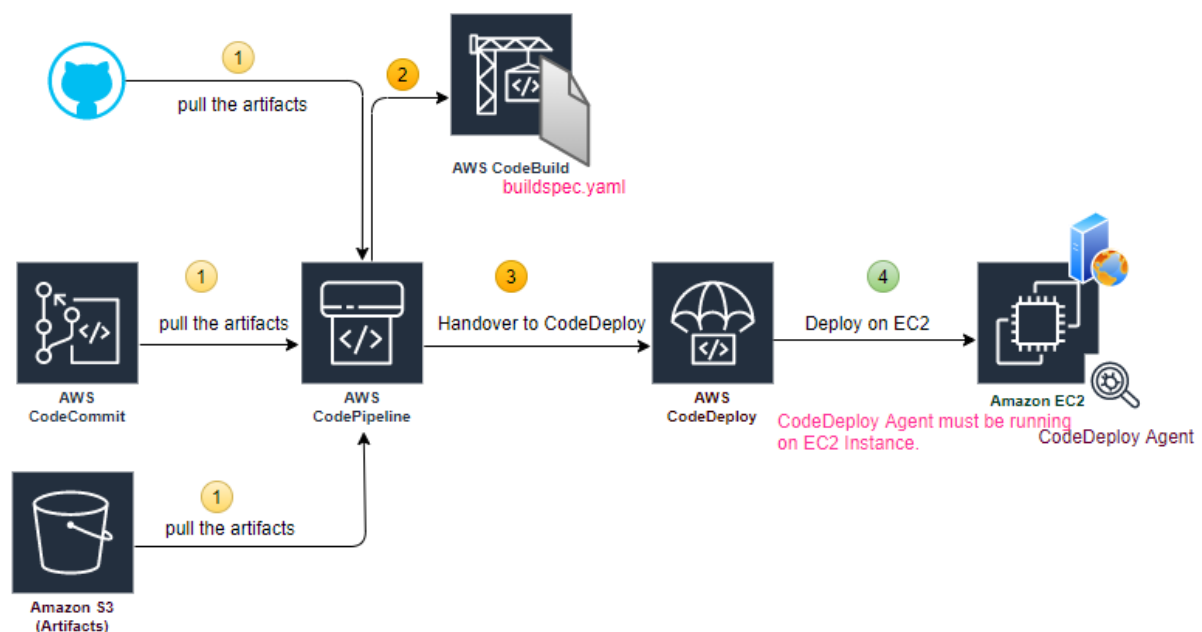
Continuous Integration happens through the CodePipeline

step1: fetch the source code from version control system (s3, Github).

step2: Then build

step3: handover the build to CodeDeploy

step4: CodeDeploy will take care to deploy at EC2 Instance.



Code build

mvn package (creates a .jar file), CodeBuild do the same thing.

```
version: 0.2

phases:
  install:
    runtime-versions:
      java: corretto8
  build:
    commands:
      - echo Build started on `date`
      - mvn test
  post_build:
    commands:
      - echo Build completed on `date`
      - mvn package
artifacts:
  files:
    - target/my-app-1.0-SNAPSHOT.jar
    - appspec.yml
discard-paths: yes
```

step1: CodeBuild->Build Project-> Create Build Project

Create build project

Project configuration

Project name

A project name must be 2 to 255 characters. It can include the letters A-Z and a-z, the numbers 0-9, and the special characters - and _.

Description - optional

Source

[Add source](#)

Source 1 - Primary

Source provider

AWS CodeCommit

No source

Amazon S3

AWS CodeCommit

GitHub

Bitbucket

GitHub Enterprise

Source

Add source

Source 1 - Primary

Source provider

AWS CodeCommit

Repository

TestRepo

Reference type

Choose the source version reference type that contains your source code.

☒ Branch

☐ Git tag

☐ Commit ID

Branch

Choose a branch that contains the code to build.

main

Commit ID - optional

Choose a commit ID. This can shorten the duration of your build.

Environment: CodeBuild asking Which Machine?

Environment

Environment image

☒ Managed image
Use an image managed by AWS CodeBuild

☐ Custom image
Specify a Docker image

Operating system

Amazon Linux 2

i The programming language runtimes are now included in the standard image of Ubuntu 18.04, which is recommended for new CodeBuild projects created in the console. See [Docker Images Provided by CodeBuild for details](#).

Runtime(s)

Standard

Image

aws/codebuild/amazonlinux2-x86_64-standard:3.0

Image version

Always use the latest image for this runtime version

Privileged

☐ Enable this flag if you want to build Docker images or want your builds to get elevated privileges

Buildspec

Build specifications

☒ **Use a buildspec file**
Store build commands in a YAML-formatted buildspec file

☐ **Insert build commands**
Store build commands as build project configuration

Buildspec name - *optional*

By default, CodeBuild looks for a file named buildspec.yml in the source code root directory. If your buildspec file uses a different name or location, enter its path from the source root here (for example, buildspec-two.yml or configuration/buildspec.yml).

Artifacts

Add artifact

Artifact 1 - Primary

Type

No artifacts ▼

You might choose no artifacts if you are running tests or pushing a Docker image to Amazon ECR.

► Additional configuration

Cache, encryption key

Logs

CloudWatch

☒ **CloudWatch logs - *optional***
Checking this option will upload build output logs to CloudWatch.

Group name

Stream name

S3

☐ **S3 logs - *optional***
Checking this option will upload build output logs to S3.

CodeBuild, instantiate a new build server provided by aws at run time and executes the instructions on buildspec.yml.

step2: Create CodePipeline for CodeBuild

step2.1: pipelines-> create pipeline

step2.2: source

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

Search:

- AWS CodeCommit
- Amazon ECR
- Amazon S3
- Bitbucket
- GitHub (Version 1)
- GitHub (Version 2)
- GitHub Enterprise Server

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step2.3: build stage

Add build stage [Info](#)

Build - optional

Build provider
This is the tool of your build project. Provide build artifact details like operating system, build spec file, and output file names.

Search:

- AWS CodeBuild
- Add Jenkins
- Custom DemoFourStageProvider (Version: 1)
- Custom DemoFromJenkins-01 (Version: 1)
- Custom DemoJenkinsProvider (Version: 1)

[Skip build stage](#) [Next](#)

step2.4: deployment stage

Add deploy stage [Info](#)

Deploy - optional

Deploy provider
Choose how you deploy to instances. Choose the provider, and then provide the configuration details for that provider.

- Q
- AWS AppConfig
- AWS CloudFormation
- AWS CloudFormation Stack Set
- AWS CodeDeploy
- AWS Elastic Beanstalk
- AWS OpsWorks Stacks
- AWS Service Catalog
- Amazon ECS
- Amazon ECS (Blue/Green)
- Amazon S3

[Skip deploy stage](#) [Next](#)

Add deploy stage [Info](#)

Deploy - optional

Deploy provider
Choose how you deploy to instances. Choose the provider, and then provide the configuration details for that provider.

AWS CodeDeploy

Region

Asia Pacific (Mumbai)

Application name
Choose an application that you have already created in the AWS CodeDeploy console. Or create an application in the AWS CodeDeploy console and then return to this task.

Q CodeDeploy-App-02 X

Deployment group
Choose a deployment group that you have already created in the AWS CodeDeploy console. Or create a deployment group in the AWS CodeDeploy console and then return to this task.

Q CodeDeploy-App-Group-02 X

step2.5: Create a CodePipeline.

Success
 Congratulations! The pipeline CodePipeline-App-03 has been created.

Create a notification rule for this pipeline

Source

AWS CodeCommit

Succeeded - Just now

072cf7ff

072cf7ff Source: Testing Commit

Disable transition

Build

In progress

Pipeline execution ID: 28f9156f-8b81-47e0-bf1c-b508f20e8ec9

Build

AWS CodeBuild

In progress - Just now

Details

072cf7ff Source: Testing Commit

click on details of build stage

Build logs	Phase details	Reports	Environment variables	Build details	Resource utilization
Name	Status	Context	Duration	Start time	End time
SUBMITTED	Succeeded	-	<1 sec	Jun 20, 2021 7:22 AM (UTC+5:30)	Jun 20, 2021 7:22 AM (UTC+5:30)
QUEUED	Succeeded	-	1 sec	Jun 20, 2021 7:22 AM (UTC+5:30)	Jun 20, 2021 7:22 AM (UTC+5:30)
PROVISIONING	Succeeded	-	19 secs	Jun 20, 2021 7:22 AM (UTC+5:30)	Jun 20, 2021 7:23 AM (UTC+5:30)
DOWNLOAD_SOURCE	Succeeded	-	1 sec	Jun 20, 2021 7:23 AM (UTC+5:30)	Jun 20, 2021 7:23 AM (UTC+5:30)
INSTALL	Succeeded	-	<1 sec	Jun 20, 2021 7:23 AM (UTC+5:30)	Jun 20, 2021 7:23 AM (UTC+5:30)
PRE_BUILD	Succeeded	-	<1 sec	Jun 20, 2021 7:23 AM (UTC+5:30)	Jun 20, 2021 7:23 AM (UTC+5:30)
BUILD	Succeeded	-	79 secs	Jun 20, 2021 7:23 AM (UTC+5:30)	Jun 20, 2021 7:24 AM (UTC+5:30)
POST_BUILD	Succeeded	-	10 secs	Jun 20, 2021 7:24 AM (UTC+5:30)	Jun 20, 2021 7:24 AM (UTC+5:30)
UPLOAD_ARTIFACTS	Succeeded	-	<1 sec	Jun 20, 2021 7:24 AM (UTC+5:30)	Jun 20, 2021 7:24 AM (UTC+5:30)
FINALIZING	Succeeded	-	<1 sec	Jun 20, 2021 7:24 AM (UTC+5:30)	Jun 20, 2021 7:24 AM (UTC+5:30)
COMPLETED	Succeeded	-	-	Jun 20, 2021 7:24 AM (UTC+5:30)	-

UPLOAD_ARTIFACTS: handover the artifacts to CodePipeline, CodePipeline handover it to CodeDeploy.

Click on CodeDeploy Events

Event	Duration	Status	Error code	Start time	End time
ApplicationStop	less than one second	✔ Succeeded	-	Jun 20, 2021 7:24 AM (UTC+5:30)	Jun 20, 2021 7:24 AM (UTC+5:30)
DownloadBundle	less than one second	✔ Succeeded	-	Jun 20, 2021 7:24 AM (UTC+5:30)	Jun 20, 2021 7:24 AM (UTC+5:30)
BeforeInstall	less than one second	✔ Succeeded	-	Jun 20, 2021 7:24 AM (UTC+5:30)	Jun 20, 2021 7:24 AM (UTC+5:30)
Install	less than one second	✔ Succeeded	-	Jun 20, 2021 7:25 AM (UTC+5:30)	Jun 20, 2021 7:25 AM (UTC+5:30)
AfterInstall	less than one second	✔ Succeeded	-	Jun 20, 2021 7:25 AM (UTC+5:30)	Jun 20, 2021 7:25 AM (UTC+5:30)
ApplicationStart	less than one second	✔ Succeeded	-	Jun 20, 2021 7:25 AM (UTC+5:30)	Jun 20, 2021 7:25 AM (UTC+5:30)
ValidateService	less than one second	✔ Succeeded	-	Jun 20, 2021 7:25 AM (UTC+5:30)	Jun 20, 2021 7:25 AM (UTC+5:30)

UC: Fail the build

In case of build fail, notify the user

Using the SNS => Simple Notification Service

step1: create a SNS topic.

step1.1: create a subscription.

Amazon SNS > Subscriptions > Create subscription

Amazon Kinesis Data Firehose

Amazon SQS

AWS Lambda

Email

Email-JSON

HTTP

HTTPS

Platform application endpoint

SMS

Select protocol

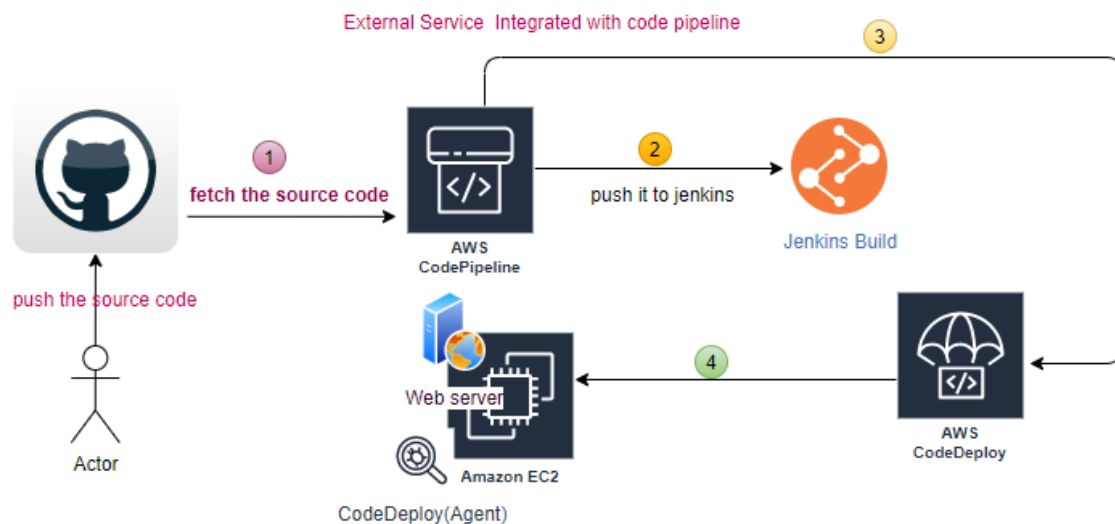
After your subscription is created, you must confirm it. Info

step1.2: confirm the subscription (email).

Build Server created and destroyed by AWS at run time.

step2: CodePipeline enables notification for us.

How to integrate External Version Control (GitHub), Build Service (Jenkins) in our CodePipeline?



AWS pipeline pulls the source code from GitLab, handover it to Jenkins. Jenkins build the source code and handover back to CodePipeline, then CodePipeline handover the artifact to the CodeDeploy and this deploy the artifact to deployment group EC2.

How the changes in GitHub automatically updated in AWS?

CloudWatch=>Events=>Rules (informed to Pipeline there is change in CodeCommit):

created-by Pipeline automatically.

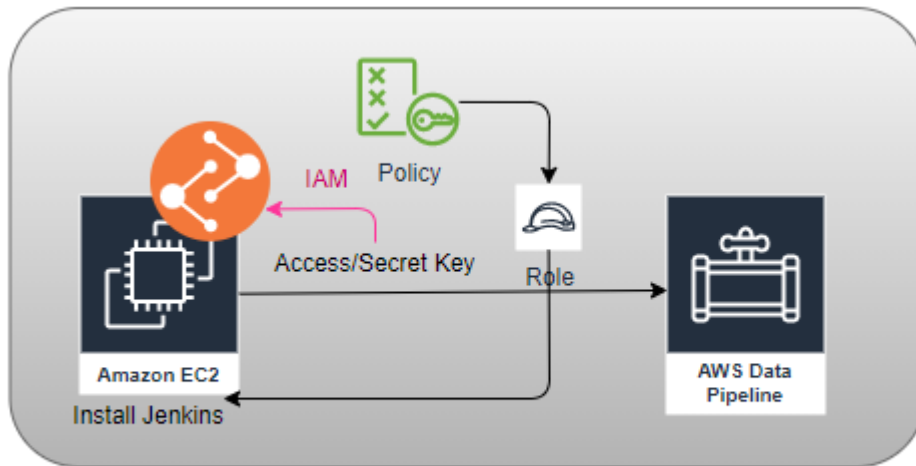
step1: source code available in GitHub.

step2: Launch Jenkins server in EC2 (SG 8080).

step2.1: EC2 is an AWS resource, requires role to interact with CodePipeline

UC: Jenkin is an external service, any external service how to communicate with AWS service?

Using Access/Secret Keys



My Account

External Service (Jenkins) don't have Role, it comes with IAM User and that IAM user must carry certain policies.

Role automatically goes to EC2, to access the AWS resource, and IAM User access key and secret key must be used by external service (Jenkins) to communicate with CodePipeline, and this must be within same account.

EC2 role=> `AWSCodePipelineCustomActionAccess`

Access/Secret Keys: Used by external services to access the AWS resources.

IAM User Policy-> `AWSCodePipelineCustomActionAccess`

In Jenkins

AWS CodePipeline Plugin

AWS CodeDeploy Plugin, AWS CodeBuild Plugin

Source Code Management

☐ None
 ☒ AWS CodePipeline

AWS Config

AWS Region

Asia Pacific (Mumbai) ap-south-1

Proxy Host

Proxy Port

0

Credentials

AWS Secret Key ?

Concealed Change Password

☒ Clear workspace before copying

CodePipeline Action Type ?

This value must match the Category field that is on the Custom Action in your corresponding Pipeline.

Category ?

Build ▼

This value must match the Provider field that is on the Custom Action in your corresponding Pipeline.

Provider ?

Demo-Jun-2021-provider

This value must match the Version field that is on the Custom Action in your corresponding Pipeline.

Version ?

Provider=> same as mention in AWS

Create CodePipeline

step3.1: source

Add source stage Info

Source

Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.

- AWS CodeCommit
- Amazon ECR
- Amazon S3
- Bitbucket
- GitHub (Version 1)
- GitHub (Version 2)**
- GitHub Enterprise Server

Previous Next

Connection: Connect to GitHub

Repository name
Choose a repository in your GitHub account.

shrikantja/aws-codepipeline-jenkins-aws-codedeploy_linux

<account>/<repository-name>

Branch name
Choose a branch of the repository.

master

Change detection options

☒ Start the pipeline on source code change
Automatically starts your pipeline when a change occurs in the source code. If turned off, your pipeline only runs if you start it manually or on a schedule.

Output artifact format

Once you select the start the pipeline option (CloudWatch is created)

Build:

step3.2: Build

Add build stage

Build - optional

Build provider
This is the tool of your build project. Provide build artifact details like operating system, build spec file, and output file names.

AWS CodeBuild

Add Jenkins

Custom DemoFourStageProvider (Version: 1)

Custom DemoFromJenkins-01 (Version: 1)

Custom DemoJenkinsProvider (Version: 1)

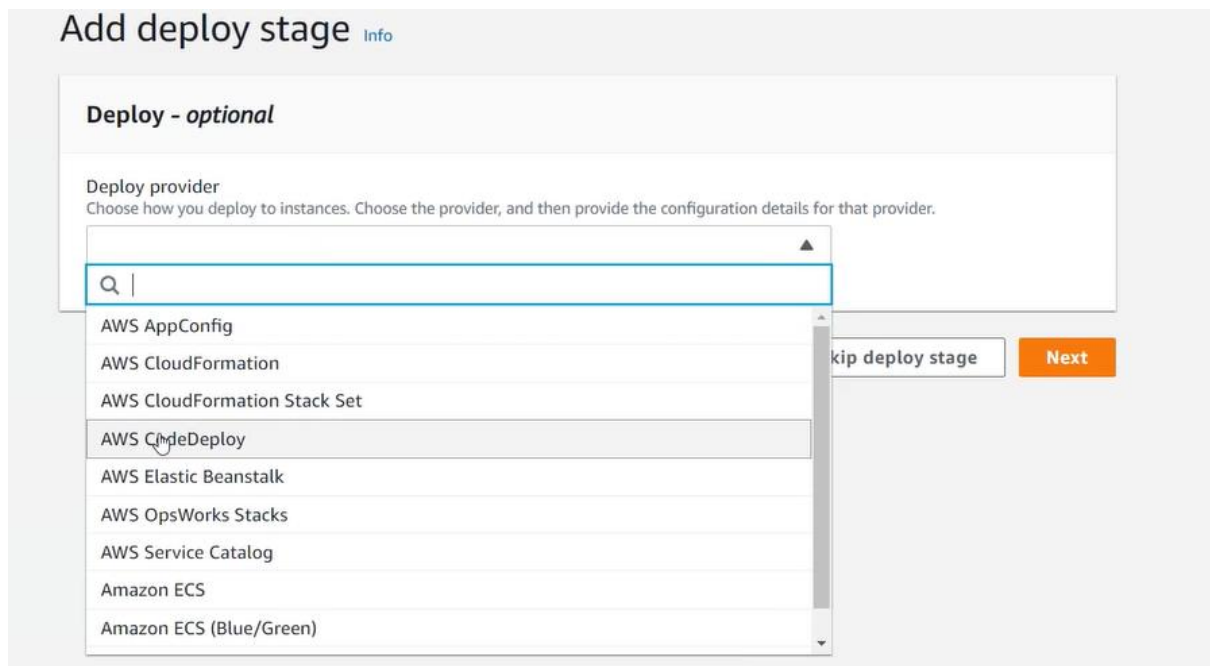
Skip build stage

Next

Provider: must be matched with provider name as in the Jenkins

Deploy

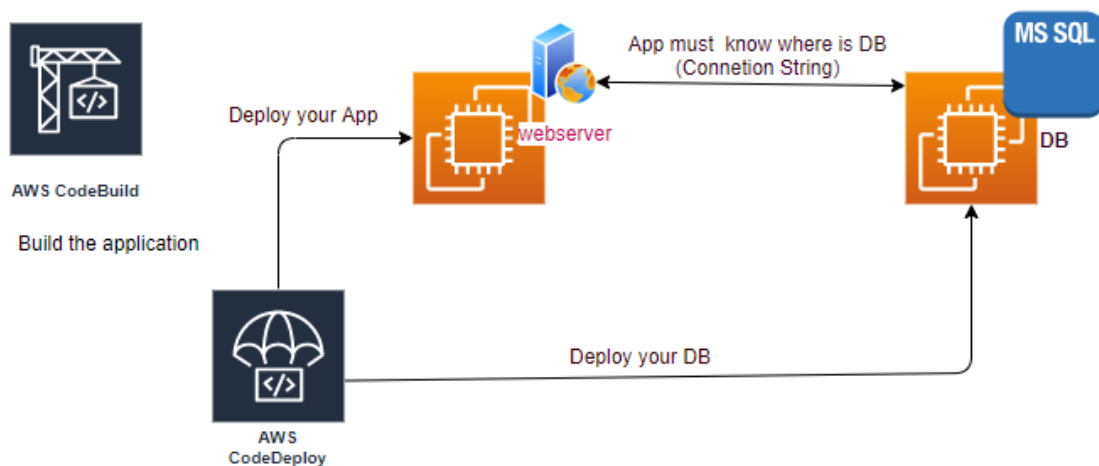
step3.3: Deploy



Source: GitHub, CodeCommit, S3

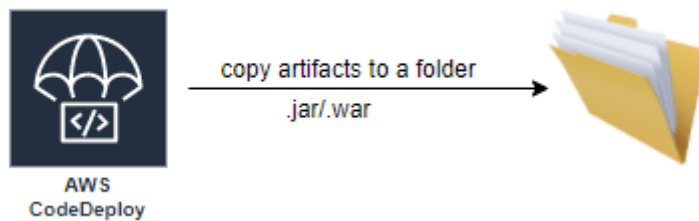
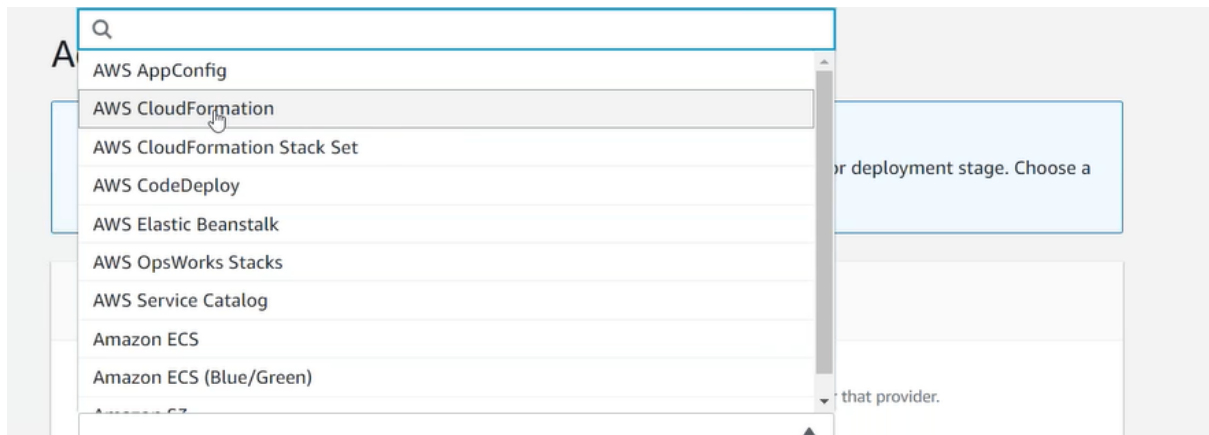
Build: Jenkins, CodeBuild

Deploy: CodeDeploy



How to define Connection String, connects both Instance, Use Configuration Management tool, e.g., Ansible/puppet/Chef. It's possible through OpsWorks.

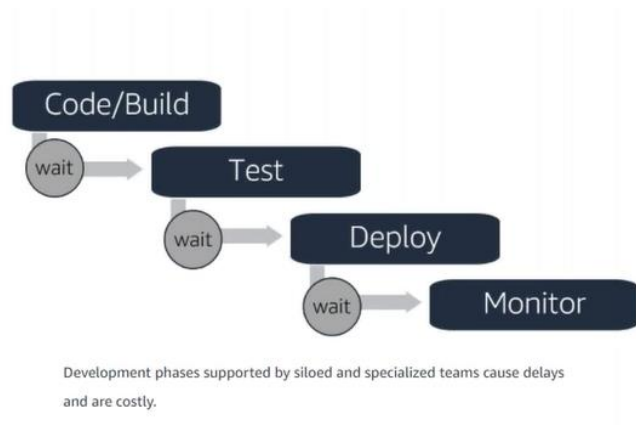
Different Deployment Stages in CodePipeline:



book: phoenix project (Gene Kim):

Waterfall:

- Waterfall development projects are slow, not iterative, resistant to change, and have long release cycles. Some reasons for this include:
 - Requirements are rigid, set at project start, and will likely not change.
 - Development phases are siloed, each starting after the previous phase has ended. Each phase is supported by highly specialized teams. Handoffs from one phase to the other are long, often requiring teams to switch tools and spend time clarifying incomplete or ambiguous information.
 - Testing and security come after implementation, making corrective actions responsive and expensive.



CodeBuild supports only two Operating Systems, if you want to use Window server then go for Jenkins.

Environment

Environment image

☒ **Managed image**
Use an image managed by AWS CodeBuild

☐ **Custom image**
Specify a Docker image

Operating system

Amazon Linux 2

Ubuntu

elevated privilege Ubuntu

DevOps is a concept (Culture), introduced in AWS through CodePipeline.

- Agility
- Rapid Delivery
- Reliability
- Scale
- Improved Collaboration
- Security

Domain	% of Examination
Domain 1: SDLC Automation	22%
Domain 2: Configuration Management and Infrastructure as Code	19%
Domain 3: Monitoring and Logging	15%
Domain 4: Policies and Standards Automation	10%
Domain 5: Incident and Event Response	18%
Domain 6: High Availability, Fault Tolerance, and Disaster Recovery	16%
TOTAL	100%

AWS Services

- List the AWS services available to implement a successful DevOps methodology.
- Identify the AWS services used to automate the CI and CD process.

